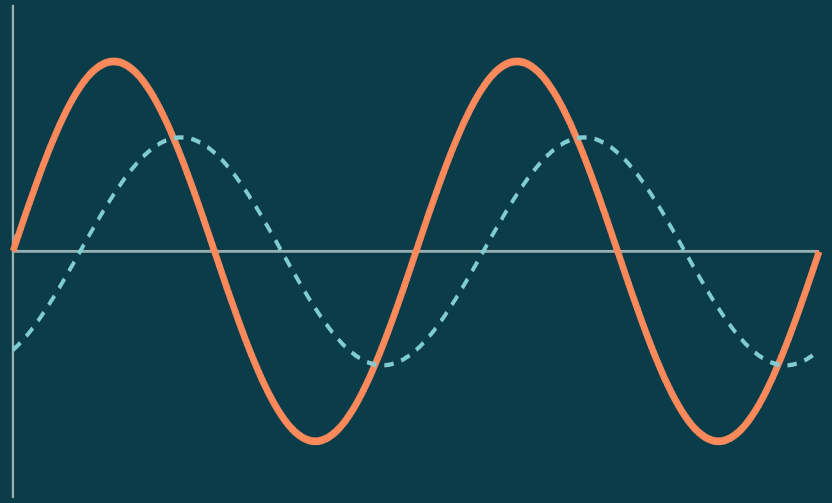
Sinusoidal AC: Values and Phasors



Basic Electrical and Electronics Engineering

Unit 1: Circuit Fundamentals

Topic: Sinusoidal quantities and phasors

Prof. Mitesh Astik

Learning Outcomes

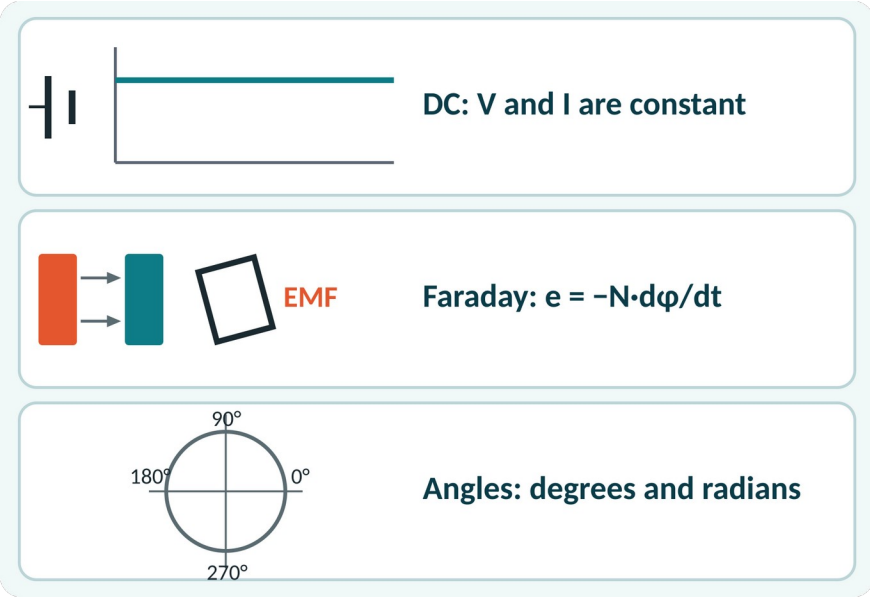
- Explain how a rotating coil generates a sinusoidal voltage.
- Calculate period, frequency, average and RMS values.
- Find form factor and peak factor of a sine wave.
- Represent and add sinusoids using phasors.



Linked to CO1
Analyse DC and AC electrical circuits to determine circuit parameters.

Recap: What You Already Know

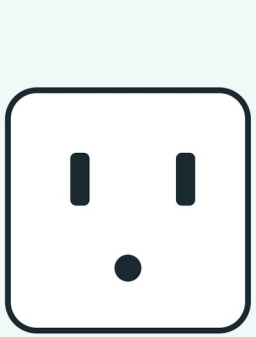
- DC: v and i are constant in time.
- Ohm's law: V = I·R; power P = V·I.
- Faraday's law: e = −N·dφ/dt.
- Trigonometry: sin, cos, radians and degrees.

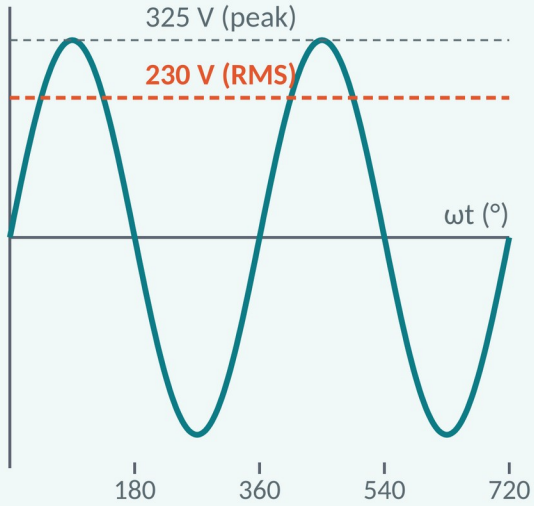


e = −N·dφ/dt
e induced EMF (V), N turns, φ flux (Wb)

Motivation: What Does “230 V” Mean?

- Wall socket: 230 V, 50 Hz.
- That 230 V is an RMS value, not peak.
- The actual peak voltage is about 325 V.
- AC is easy to step up and down with transformers.

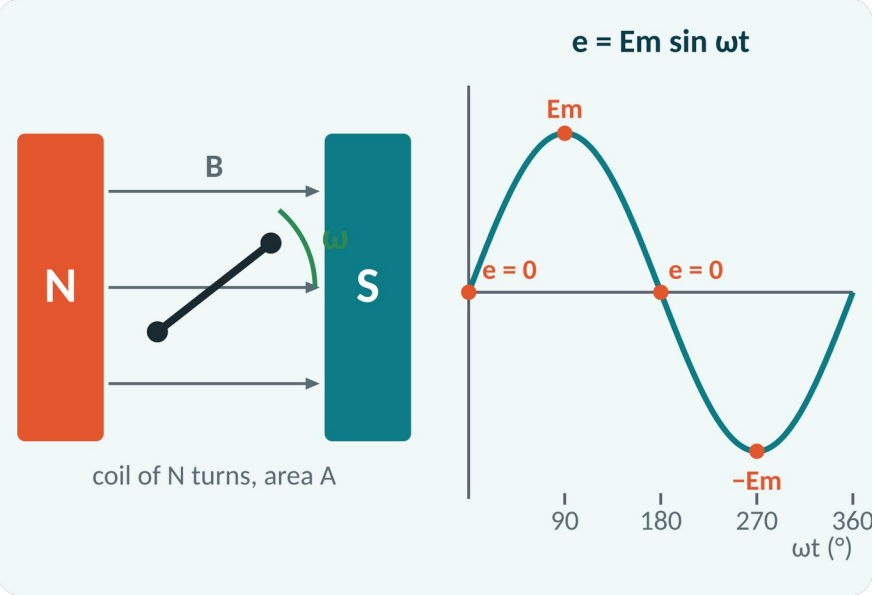




$$V_m = \sqrt{2} \cdot V = 1.414 \times 230 \approx 325 \text{ V}$$

Generation of Sinusoidal Voltage

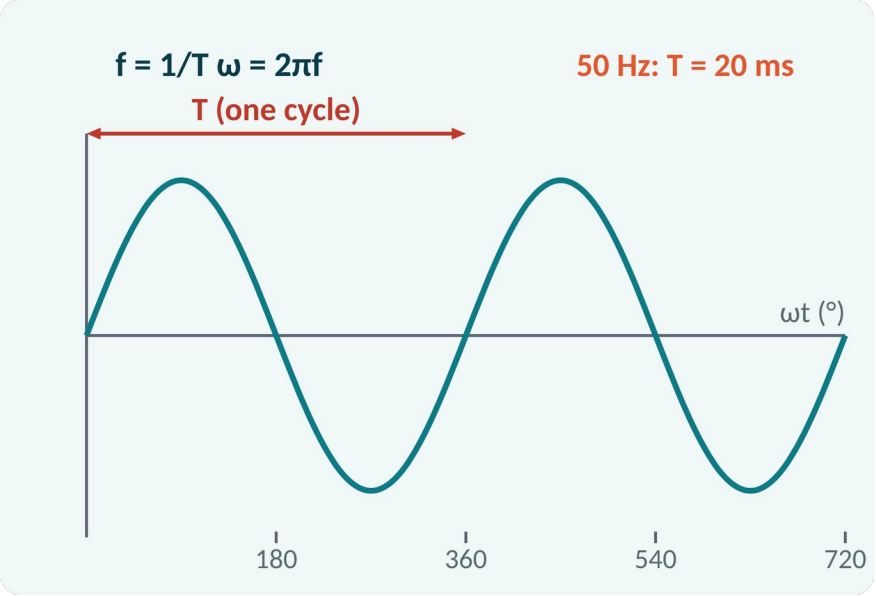
- A coil of N turns rotates in a uniform field.
- Flux linked: φ = Φm cos ωt.
- EMF is zero when coil sides move along the field.
- EMF is maximum when coil sides cut the field.
- Slip rings and brushes carry the voltage out.



e = Em sin ωt, Em = N·B·A·ω
N turns, B flux density (T), A coil area (m²), ω speed (rad/s)

Basic Terms: Cycle, Period, Frequency

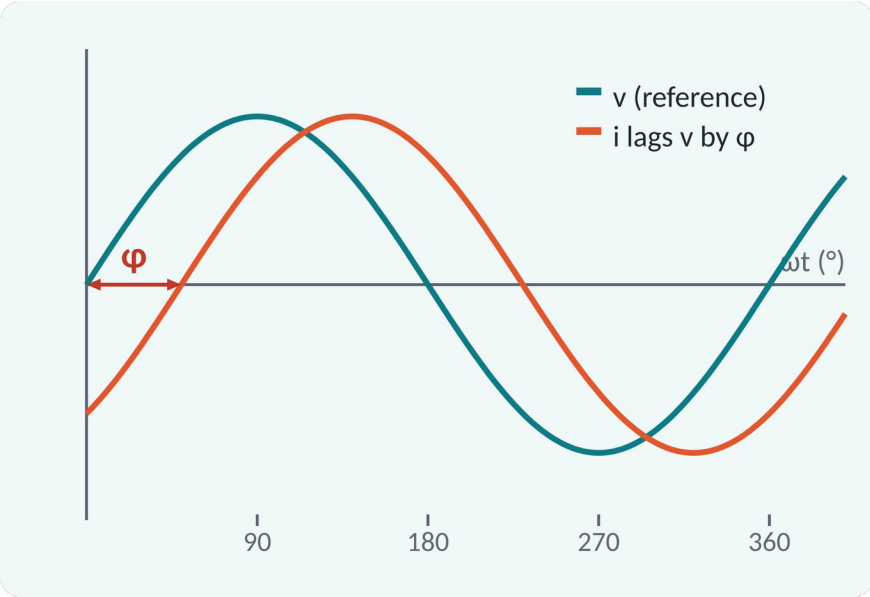
- Cycle: one complete set of positive and negative values.
- Period T: time for one cycle (s).
- Frequency f: cycles per second (Hz).
- Angular frequency ω: radians per second.
- Generator: f = P·Ns/120 (P poles, Ns rpm).



f = 1/T; ω = 2πf
For 50 Hz: T = 20 ms, ω = 314.16 rad/s

Equation of a Sinusoid and Phase

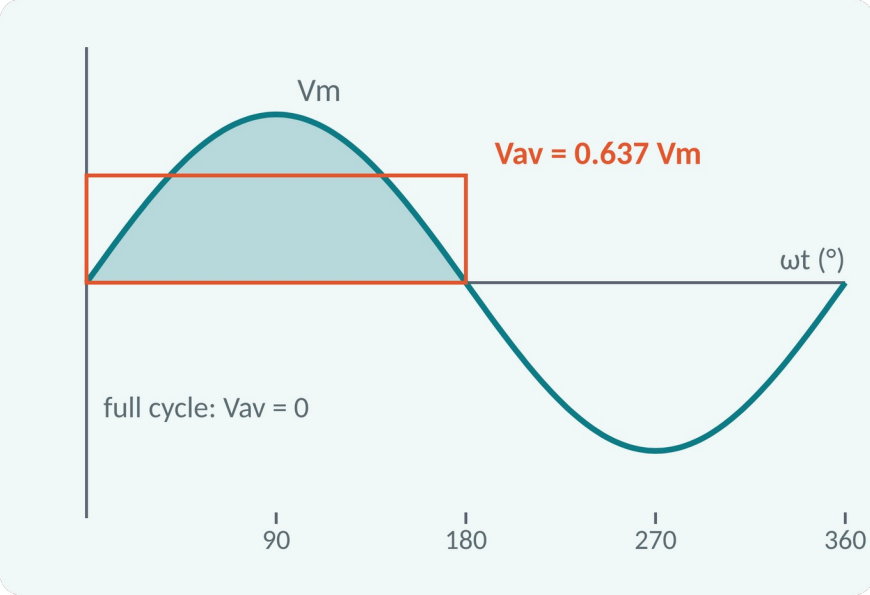
- General form: v = Vm sin(ωt + φ).
- φ positive: wave leads the reference.
- φ negative: wave lags the reference.
- 0° in phase; 90° quadrature; 180° opposition.



v = Vm sin(ωt + φ), i = Im sin(ωt − φ)
Vm peak (V), φ phase angle (rad or °)

Average Value

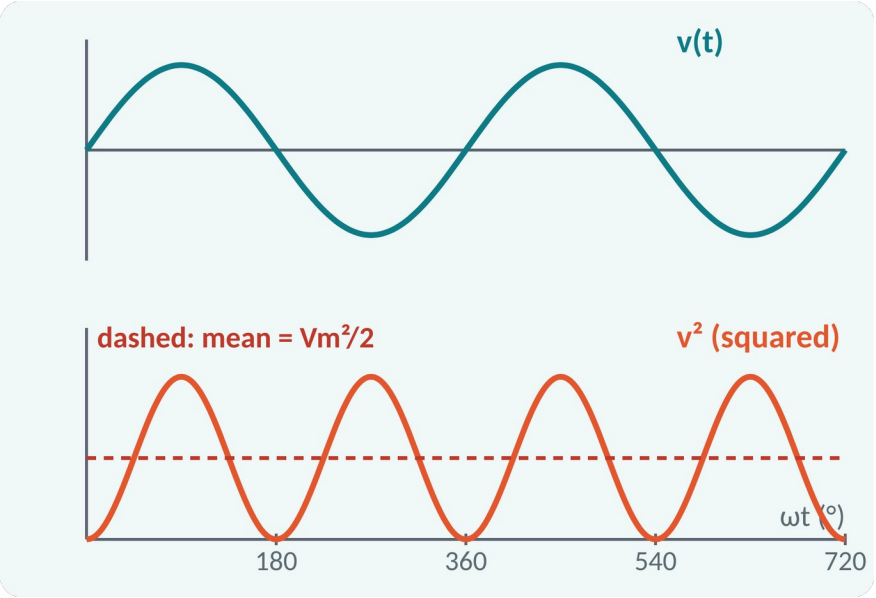
- Average of a full sine cycle is zero.
- So it is defined over a half cycle.
- Equals the height of a rectangle of equal area.
- Used for rectified signals and DC meters.



Vav = (1/T)∫v dt; half cycle: Vav = (1/π)∫₀^π Vm sin θ dθ
Vav = 2Vm/π = 0.637·Vm

RMS (Effective) Value

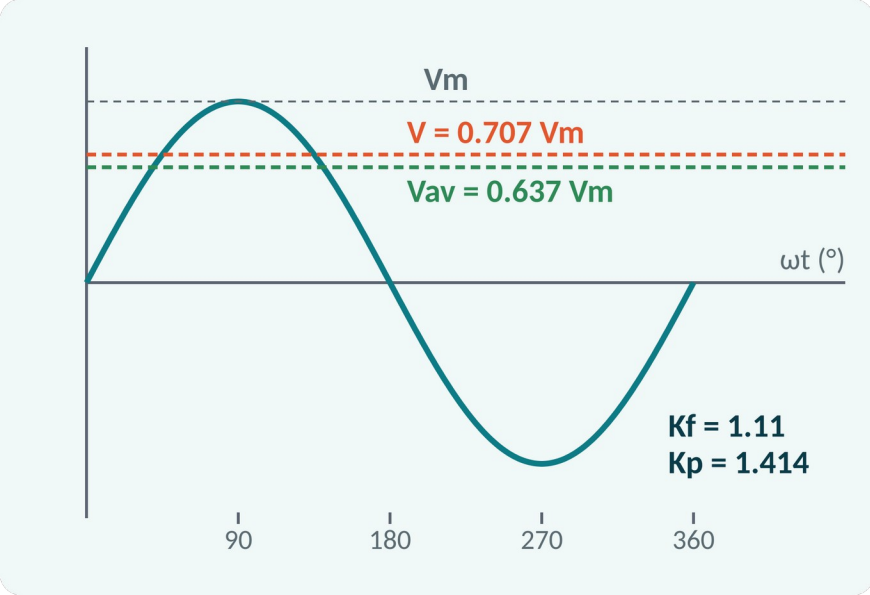
- RMS: root of the mean of the squares.
- Equals the DC value giving the same heating.
- For a sine, v^2 has mean $Vm^2/2$.
- Meters and ratings use the RMS value.



$V = \sqrt{[(1/T)\int v^2\,dt]}$; sine: $V = Vm/\sqrt{2} = 0.707 \cdot Vm$
Identity: $\sin^2\theta = (1 - \cos 2\theta)/2$

Form Factor and Peak Factor

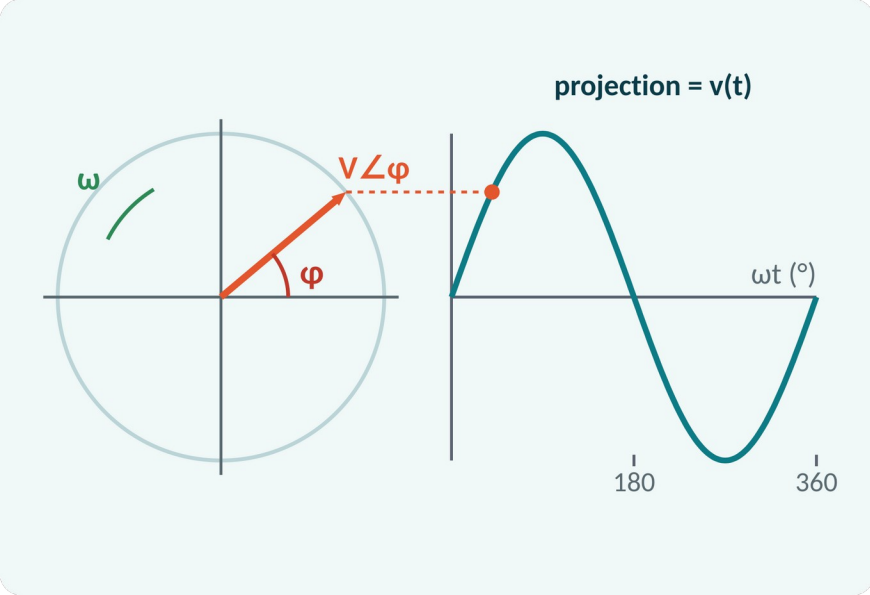
- Form factor = RMS value / average value.
- Peak factor = peak value / RMS value.
- They describe the shape of a waveform.
- Sine wave: Kf = 1.11, Kp = 1.414.



Kf = V/Vav = 1.11; Kp = Vm/V = √2 = 1.414
Sine: V = 0.707 Vm, Vav = 0.637 Vm (valid for sine only)

Phasor Representation

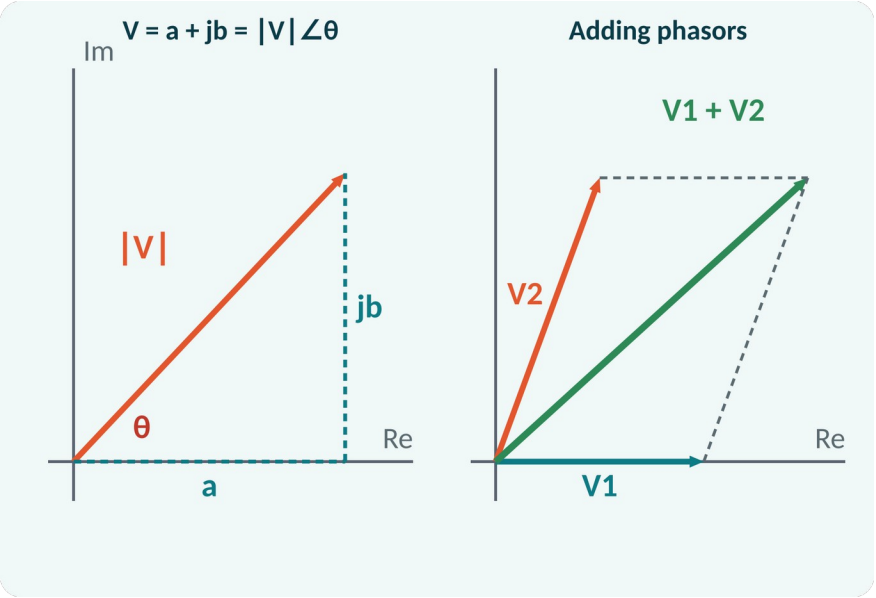
- A phasor is a rotating line for a sinusoid.
- Length = RMS value; angle = phase angle.
- Vertical projection gives the instantaneous value.
- Same-frequency waves keep a fixed angle.



v = Vm sin(ωt + φ) ⇄ V = V∠φ
V RMS magnitude (V), φ phase angle (°)

Working with Phasors

- Rectangular form: V = a + jb.
- Polar form: V = |V|∠θ.
- Add and subtract in rectangular form.
- Multiply and divide in polar form.
- Phasors add only if frequencies are equal.

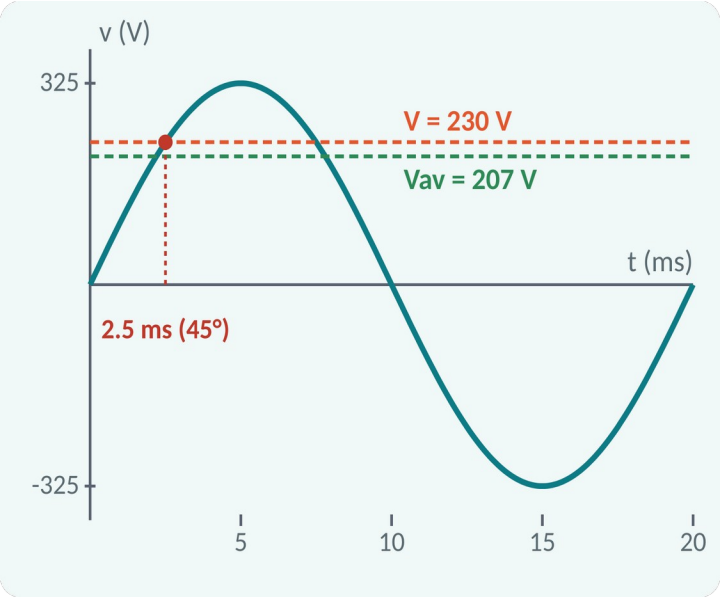


$|V| = \sqrt{(a^2 + b^2)}; \quad \theta = \tan^{-1}(b/a)$

$a = |V|\cos\theta; \quad b = |V|\sin\theta$

Solved Example 1: Sine Wave Values

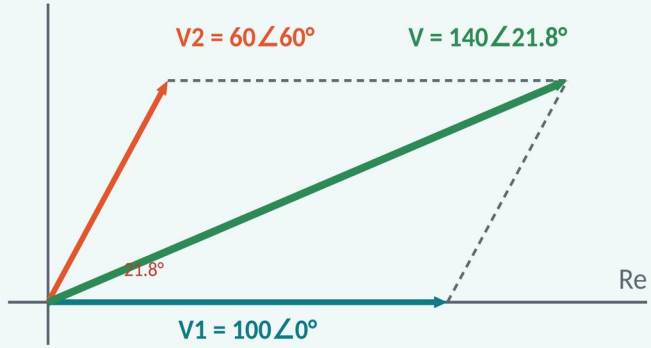
- Given v = 325 sin(314t) V. Find Vm, V, f, T, Vav and v at 2.5 ms.
- 1. Vm = 325 V.
- 2. V = Vm/√2 = 325/1.414 ≈ 230 V.
- 3. f = ω/2π = 314/6.283 ≈ 50 Hz.
- 4. T = 1/f = 20 ms.
- 5. Vav = 0.637 × 325 ≈ 207 V (half cycle).
- 6. ωt = 314 × 0.0025 = 0.785 rad = 45°.
- 7. v = 325 sin 45° ≈ 230 V.



Answer: Vm = 325 V, V ≈ 230 V, f = 50 Hz, T = 20 ms, Vav ≈ 207 V, v(2.5 ms) ≈ 230 V

Solved Example 2: Phasor Addition

- Given v1 = 141.4 sin ωt, v2 = 84.9 sin(ωt + 60°) V. Find v1 + v2.
- 1. RMS phasors: V1 = 100∠0°, V2 = 60∠60° V.
- 2. V2 = 60 cos 60° + j60 sin 60° = 30 + j51.96.
- 3. V = V1 + V2 = 130 + j51.96 V.
- 4. $|V| = \sqrt{(130^2 + 51.96^2)} = \sqrt{19600} = 140$ V.
- 5. $\theta = \tan^{-1}(51.96/130) = 21.8°$.
- 6. Vm = 140 × √2 = 198 V.



Answer: v = 198 sin(ωt + 21.8°) V (V = 140∠21.8° V RMS)

Common Mistakes and Exam Tips

Common mistakes

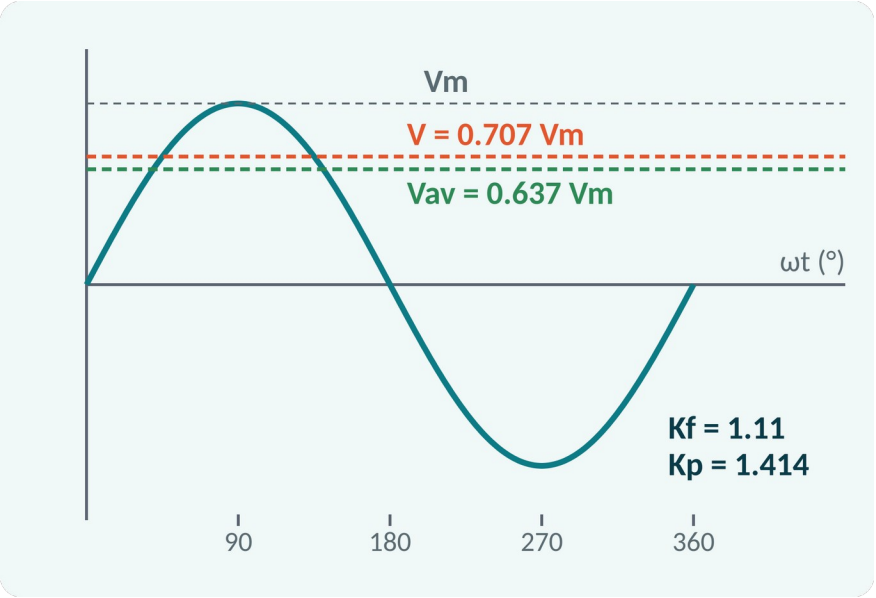
- Mixing up peak and RMS values.
- Calculator in degree mode for ωt in radians.
- Using 0.637·Vm for a full cycle (it is 0).
- Applying 1.11 and 1.414 to non-sine waves.
- Adding phasor magnitudes directly.

Exam tips

- Check whether the given value is peak or RMS.
- Meters and ratings show RMS values.
- Half-wave rectified sine: Vav = Vm/π, V = Vm/2, Kf = 1.57, Kp = 2.
- Write given, formula, substitution, answer, unit.

Summary

- Rotating coil: e = Em sin ωt, Em = N·B·A·ω.
- f = 1/T, ω = 2πf; mains is 50 Hz.
- Vav = 0.637 Vm (half cycle); V = 0.707 Vm.
- Kf = 1.11 and Kp = 1.414 for a sine.
- Phasor V = V∠φ; add in rectangular form.



Remember four numbers: 0.637, 0.707, 1.11, 1.414

Quick Quiz

Q1. Mains with 325 V peak has RMS value:
(a) 207 V (b) 230 V (c) 325 V (d) 460 V

Ans: (b)

Q2. Form factor of a sine wave is:
(a) 1.0 (b) 1.11 (c) 1.414 (d) 1.57

Ans: (b)

Q3. Period T = 25 ms. Frequency is:
(a) 25 Hz (b) 40 Hz (c) 50 Hz (d) 400 Hz

Ans: (b)

Q4. $v = 141.4 \sin(\omega t + 45°)$ V has RMS phasor:
(a) $141.4\angle 45°$ (b) $100\angle 45°$ (c) $100\angle -45°$ (d) $70.7\angle 45°$

Ans: (b)

Q5. V_m = 100 V; half-cycle average value is:
(a) 31.8 V (b) 50 V (c) 63.7 V (d) 70.7 V

Ans: (c)

References

1. **Hughes Edward, Electrical Technology**
   10th Ed., Pearson Education, 2010 (Unit 1 textbook)

2. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3–5 textbook)

3. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

4. **David A. Bell, Operational Amplifier and Linear ICs**
   3rd Ed., Oxford University Press, 2011 (reference, Unit 5)

Next lecture: R, L, C and series RL, RC, RLC circuits; power and power factor